

UTTARAKHAND CLIMATE VULNERABILITY ASSESSMENT

Report

Acknowledgement

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The climate foundation on which this work rests downscaled CMIP6 projections and extreme-climate indices at 5 km × 5 km resolution was developed by the Indian Institute of Technology Roorkee under CAFRI II. The assessment team is grateful for that contribution, which made it possible to build sectoral analysis on a consistent scientific base rather than duplicating climate modelling effort.

The assessment drew on data, review and institutional guidance from a wide set of State departments and agencies. The team gratefully acknowledges the Department of Agriculture, Department of Horticulture, Department of Animal Husbandry & Dairy Development, Uttarakhand State Rural Livelihood Mission, Forest Department, State Biodiversity Board, Department of Health & Family Welfare and State Health Mission, Uttarakhand Power Corporation Limited, Power Transmission Corporation of Uttarakhand Limited, Uttarakhand Jal Vidyut Nigam, Public Works Department, Urban Development Department, Town & Country Planning Department, Irrigation Department, Uttarakhand Jal Sansthan, Uttarakhand Peyjal Nigam, Jal Jeevan Mission, Swajal, and the Uttarakhand Tourism Development Board.

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Finally, the team records its appreciation for the officials, technical staff and community members who participated in departmental consultations, expert working group sessions and field validation exercises. Their scrutiny corrected several early assumptions and materially improved the credibility of the results presented here.

Executive Summary

This report presents the findings of the Uttarakhand Climate Vulnerability Assessment, conducted under GIZ/CAFRI II in support of the State Environment Conservation and Climate Change Directorate. It covers all 95 development blocks across the state's 13 districts, and assesses climate-induced hazard, exposure, impact, vulnerability and risk for five priority sectors: Agriculture and Allied Sectors, Water Resources and WASH, Roads and Linear Township, Health, and Energy, within a wider frame that also considers Forest and Biodiversity and Tourism.

Approach. The assessment follows the IPCC AR6 risk framing, in which risk arises from the interaction of hazard, exposure and vulnerability, and vulnerability is itself a function of sensitivity and adaptive capacity. Rather than duplicating climate modelling, it builds on the Climate Hazard Profile prepared by IIT Roorkee — downscaled CMIP6 projections and ETCCDI extreme-climate indices at 5 km × 5 km resolution under SSP2-4.5 and SSP5-8.5, for a 1981–2014 baseline and mid-century (2021–2050) and end-century (2071–2100) horizons. Eight climate-induced hazards were mapped: landslide, flood, extreme rainfall, drought, forest fire, heat stress, potentially dangerous glacial lakes, and the recorded disaster-event record. Around 120 indicators across the four risk components were assembled to produce five sectoral vulnerability indices and a Composite Vulnerability Index, each classified into six bands from Very Low to Extremely High.

A changing hazard profile. The hazard picture is not uniform, and this matters for how adaptation is targeted. Landslide and forest-fire susceptibility concentrate in the central and western mid-hill belt; flood hazard concentrates in the Tarai-Bhabar plains of Haridwar and Udham Singh Nagar; drought probability and heat stress are most pronounced across the northern high-altitude districts of Uttarkashi, Chamoli and Pithoragarh; and glacial lake exposure is confined to the high Himalayan headwaters. The recorded event record ranks Chamoli, Uttarkashi and Rudraprayag as the most disaster-affected districts, and the seasonal signature is unambiguous: the great majority of events fall within the June-to-September monsoon, peaking sharply in August.

Sectoral findings. In agriculture, aggregate productivity has improved by about 37 per cent since 2011–12, but land under traditional hill foodgrains has contracted by 27 per cent in a decade, and chilling-hour deficits are pushing orchard belts upslope - vulnerability is concentrated in rain-fed hill systems rather than in aggregate output. In water, the binding constraint is the source rather than the pipe: ninety per cent of the hill and high-hill population depends on springs fed by shallow fractured-rock aquifers, while warming advances snowmelt so that runoff arrives before, not during, the period of peak demand. In roads, a network that has more than tripled since state formation now carries structurally exposed assets across terrain that is 86 per cent mountainous, with about 23.5 per cent of inventoried bridges already 40 years or older and no practical modal substitute when a corridor fails. In health, warming at nearly twice the global rate since 1990 has opened highland microclimates to vectors previously excluded — malaria has shifted 300 metres upslope since 2000 and dengue in Dehradun rose more than thirteen-fold between 2016 and 2022. In energy, an endowment that is a climate-governed flux rather than a geological stock leaves 46.7 per cent of installed capacity in hydropower, with unserved energy projected to reach about 9.1 BU by 2035–36 on existing and planned contracts.

Convergent hotspots. Across sectors, a consistent set of blocks carries compounding exposure. High-altitude blocks in Uttarkashi, Chamoli, Rudraprayag and Pithoragarh with Bhatwari, Mori, Purola, Jyotirmath, Dasholi, Ukhimath, Munsyari and Dharchula among them combine steep, seismically active and landslide-prone terrain with glacier-fed hydrology, thin institutional presence and single-corridor dependence. The same terrain that generates landslide risk for roads destabilises hydropower assets, isolates health facilities and undermines orchard and transhumance livelihoods. These blocks warrant convergent, first-tranche investment rather than sector-siloed response.

A structural gap in what institutions are built for. Several of the most consequential hazards are chronic and slow-onset — spring depletion, chilling-hour decline, vector-range expansion, sediment loading — rather than acute. Institutional capacity in the state is oriented predominantly towards sudden-onset disaster response. Closing this gap, by treating slow-onset climate stress as a standing planning input rather than an emergency, is among the clearest priorities emerging from this assessment.

Outputs and use. The assessment delivers a block-level Digital Vulnerability Atlas navigable from state to block scale, five sectoral vulnerability indices and a composite index, block-wise risk profiles for current and future horizons, and sector-wise adaptation and mitigation measures with co-benefits, each mapped to lead departments. The intended use is practical: to inform State, district and block development plans, to enable convergence with existing schemes rather than the creation of parallel ones, and to establish a baseline against which reductions in vulnerability can be tracked over time.

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1. About the Assessment

Uttarakhand is among the most climate-exposed states in India. More than 86 per cent of its area is mountainous, over 45 per cent is under forest, and its water and energy systems depend on a cryospheric flux rather than a stored resource. Recurrent flash floods, landslides, cloudbursts, forest fires and glacial lake outburst floods have repeatedly reversed development gains — the 2013 Kedarnath floods claimed around 4,000 lives, the 2016 forest fires destroyed over 4,500 hectares, and the 2021 Chamoli rock-ice avalanche demonstrated how a single cascading event can disable major infrastructure within minutes.

This assessment, commissioned under GIZ/CAFRI II in support of the State Environment Conservation and Climate Change Directorate (SECCCD), translates that exposure into decision-ready evidence at the level of individual development blocks. It builds on the Climate Hazard Profile developed with IIT Roorkee, which provides downscaled historical and projected precipitation, temperature and extreme-climate indices at 5 km × 5 km resolution under SSP2-4.5 and SSP5-8.5, and extends it through sectoral impact assessment, vulnerability indexing, risk profiling and a Digital Vulnerability Atlas.

Uttarakhand Climate Vulnerability Assessment — At a Glance

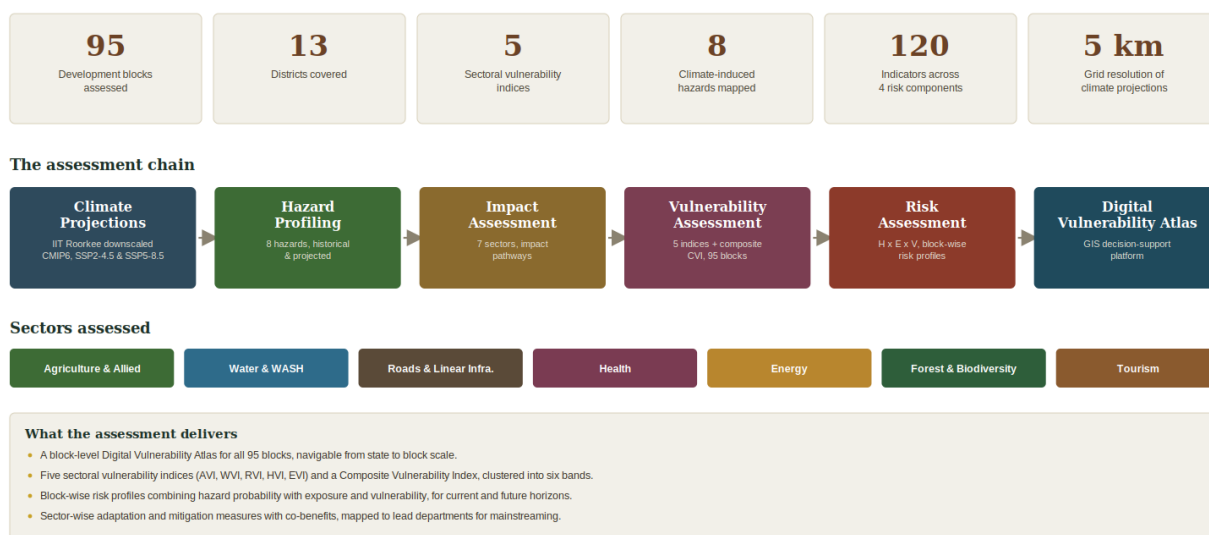


Figure 1: The Uttarakhand Climate Vulnerability Assessment at a glance — scale, assessment chain, sectors covered and intended outputs.

What this report contains

- A consolidated account of the approach: how hazard, exposure, impact, vulnerability and risk were assessed and how they connect
- A synthesis of how Uttarakhand's climate hazard profile is changing, drawing on eight mapped hazard layers and the recorded event record
- Sector chapters for Agriculture, Water & WASH, Roads & Linear Township, Health and Energy, each presenting key findings, an impact illustration, the hazard-to-risk pathway, and adaptation observations.
- A way forward setting out what worked, what proved difficult, what was learned and where the opportunities now lie

2. Approach to Developing the Vulnerability Assessment

The assessment is structured around the IPCC AR6 conception of climate risk, in which risk emerges from the interaction of a hazard with exposed people, assets and systems, and with the vulnerability of those systems. Vulnerability is treated as a function of sensitivity, how strongly a system is affected once exposed, and adaptive capacity, the ability to anticipate, absorb and recover. This framing is applied consistently across all sectors so that results remain comparable.

The five assessments and how they connect

Hazard assessment. Eight climate-induced hazards were mapped for the state: landslide, flood, extreme rainfall, drought, forest fire, heat stress, potentially dangerous glacial lakes, and the historical disaster-event record. Each was characterised by nature, geographical distribution, severity and frequency, using scientific datasets, existing hazard maps, remote sensing and the recorded event database, with deep-learning susceptibility modelling applied where conditioning-factor data supported it.

Exposure assessment. For each sector, the elements present in the hazard zone were identified and quantified — population and settlements, physical assets such as roads, transmission lines and health facilities, natural resources including cultivated and forest area, and the economic activity dependent on them.

Impact assessment. Sector-specific impact pathways were developed to trace the causal chain from a climate stressor acting on exposure, through direct impacts, to functional impacts and the disruption of services. This produced documented pathways rather than assertions, and made explicit where a single stressor propagates across sectors.

Vulnerability assessment. Around 120 indicators were selected on the basis of analytical soundness, measurability, spatial coverage and relevance, drawing on frameworks already applied in the Indian Himalayan Region. Indicators were normalised, oriented so that higher values consistently denote greater vulnerability, aggregated within components, and combined into five sectoral indices- AVI, WVI, RVI, HVI and EVI and a Composite Vulnerability Index, classified into six bands.

Risk assessment. Hazard probability was combined with exposure and vulnerability to produce block-wise risk profiles, a risk matrix and the identification of risk hotspots, for both current conditions and future horizons.

Climate Risk Framework (IPCC AR6) as applied in this assessment



Figure 2: The IPCC AR6 climate risk framework as operationalised in this assessment, showing the four risk components, the derivation of the vulnerability index, and the six-band classification applied to every block.

Methodological workflow

The workflow proceeds in six linked stages, each producing an output that becomes an input to the next. Because the Climate Hazard Profile already existed, the assessment deliberately began from that foundation rather than re-running climate models, a decision that concentrated effort on integration, sectoral analysis and the production of decision-ready outputs.

Methodological Workflow — Uttarakhand Climate Vulnerability Assessment

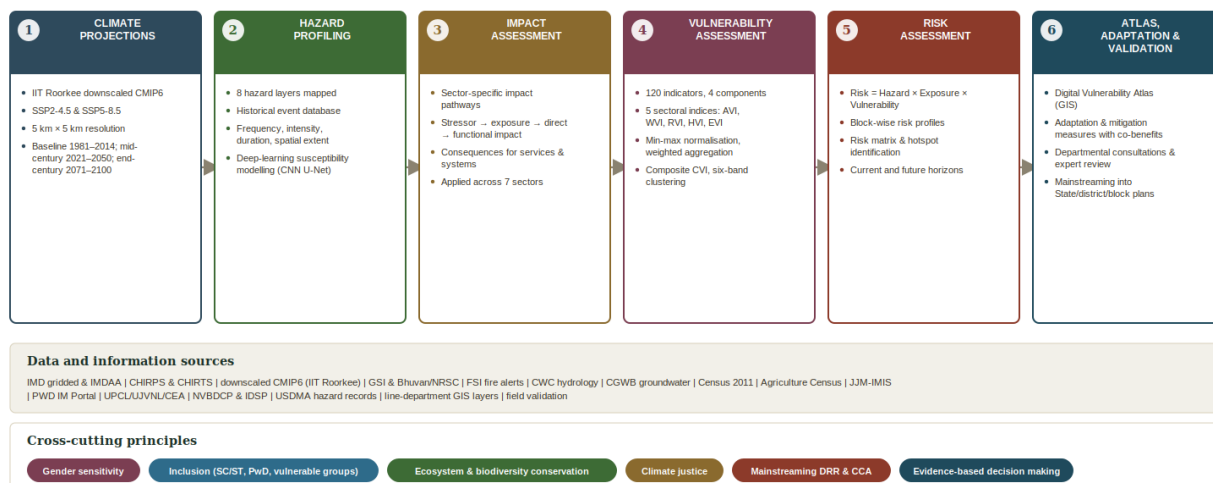
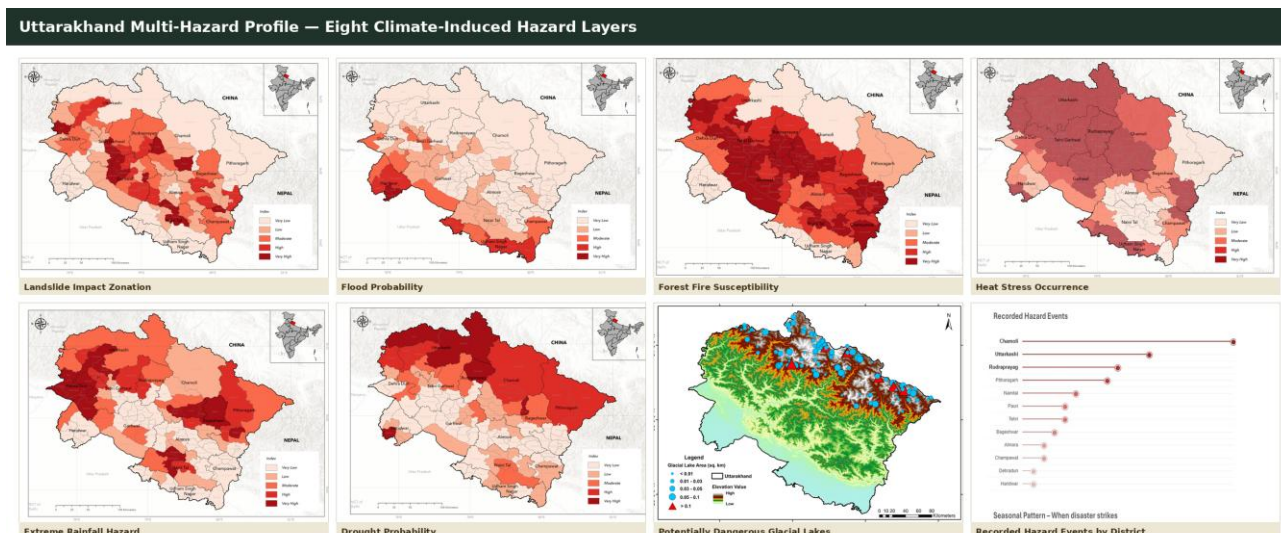


Figure 3: Methodological workflow, data and information sources, and cross-cutting principles applied throughout the assessment.

Two features of the workflow deserve emphasis. First, the treatment of indicators is transparent by design: normalisation, direction of influence and weighting are documented for every indicator, so that a result can be traced back to the data that produced it. Second, validation is not a terminal step but a recurring one, departmental consultations and expert working group review were used at successive stages, and several early indicator choices were revised as a result.

3. Changes to Uttarakhand's Climate Hazard Profile

Uttarakhand's hazard profile is changing in ways that are both measurable and spatially uneven. The eight hazard layers below were mapped at block resolution and, taken together, show that no single hazard characterises the state and that the districts facing the most severe compound exposure are not always those with the highest score on any individual layer.



Uttarakhand multi-hazard profile: climate-induced hazard layers mapped for the assessment, together with the recorded hazard event record by district.

Summary of changes to climate hazards across Uttarakhand

Warming is accelerating and is not uniform. Regional projections indicate a temperature rise of 1.3–1.6 °C by mid-century and 2.3–5.9 °C by end-century, with the state's average temperature already rising since 1990 at nearly twice the global rate. Monthly hottest-day temperatures under SSP5-8.5 rise most sharply in the high-altitude districts, with Uttarkashi, Rudraprayag and Chamoli showing changes of the order of 4.2–4.8 °C — a pattern that stresses traditional temperate crops, advances snowmelt and raises heat exposure in communities with no history of heat adaptation.

Precipitation is intensifying while becoming less reliable. Projections indicate a decline in pre-monsoon rainfall alongside a 6–16 per cent increase in monsoon precipitation by end-century, with greater year-to-year variability. Every district shows a substantial end-century increase in total precipitation — of the order of 56–63 per cent under high-emission scenarios- but the increase arrives concentrated in fewer, heavier events. The operational consequence is a simultaneous rise in flood, landslide and erosion hazard and in dry-spell and drought hazard, affecting the same blocks in different seasons.

The cryosphere is contracting. Roughly 1,950 glaciers in the state lost about 16 gigatons of ice between 2000 and 2014, some 3.5 per cent of total glacier volume. Glacier melt already contributes 10–20 per cent of annual runoff. Continued loss raises peak monsoon flows, advances the timing of snowmelt runoff, and expands the inventory of potentially dangerous glacial lakes concentrated in the high Himalayan headwaters of Chamoli, Uttarkashi and Pithoragarh.

Landslide hazard is rising and is amplified by land use. More than half the state's area is classified as highly or very highly susceptible to landslides, and in some districts climate change could raise monsoon-season landslide risk by as much as 90 per cent in the latter half of the century. The mapped pattern concentrates in the western and central mid-hill belt, and evidence from the road sector indicates that construction on cut slopes is itself a driver of destabilisation rather than only a casualty of it.

Forest fire has become a recurrent, expanding hazard. With nearly 45 per cent of the state under forest and two-thirds of that area highly fire-prone, 11,256 forest fire incidents were recorded between November 2023 and June 2024 across 11 of the 13 districts. Rising temperatures combined with declining pre-monsoon rainfall lengthen and

intensify the fire season, and the mapped susceptibility is highest across the central and southern mountainous belts.

The seasonal and spatial signature is consistent. The recorded event database shows Chamoli, Uttarkashi and Rudraprayag as the most affected districts and Haridwar and Dehradun as the least, with a pronounced monsoon concentration: the great majority of recorded events occur between June and September, peaking in August. This concentration means that hazards which are analytically separate arrive together in practice, and that adaptation measures competing for the same pre-monsoon implementation window need explicit sequencing.

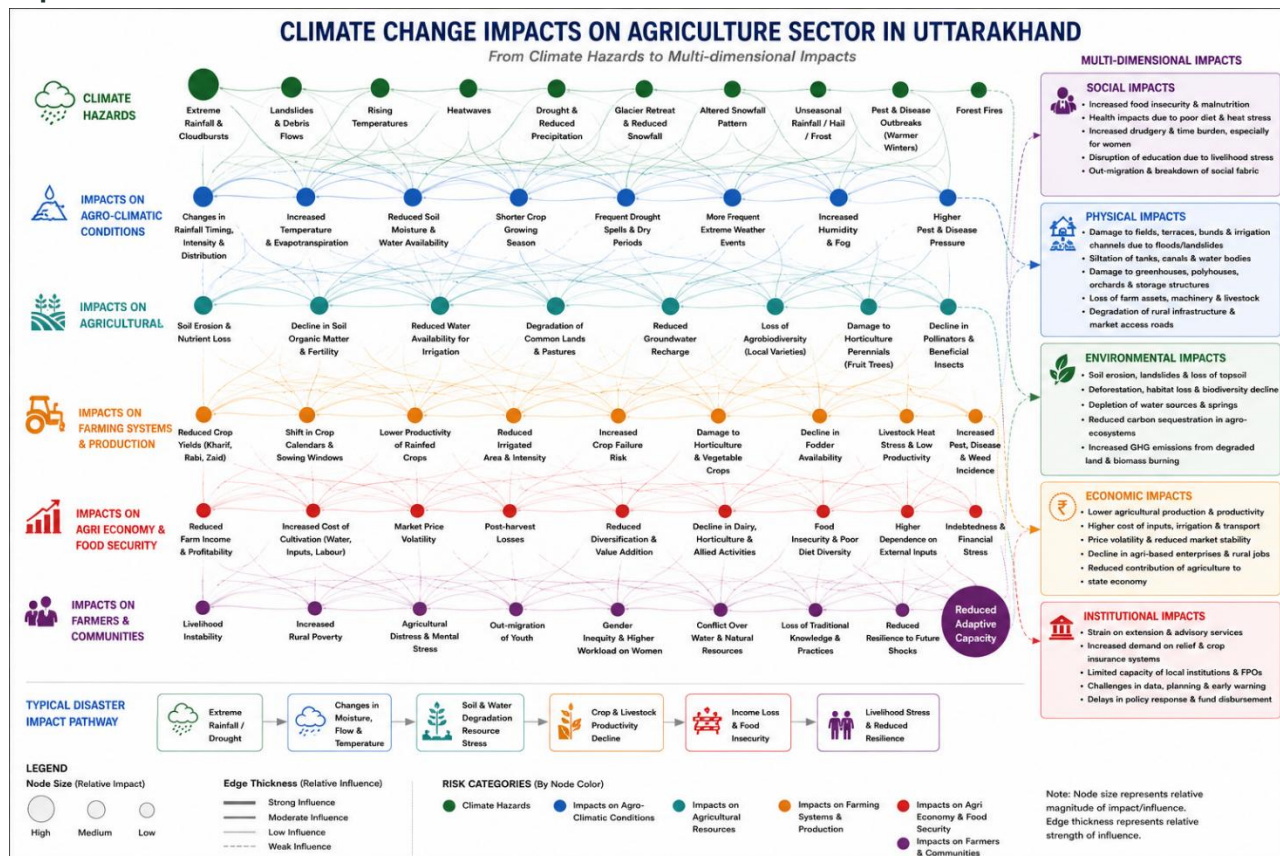
4. Agriculture & Allied Sectors

Key points

Agriculture supports the livelihoods of nearly 70 per cent of Uttarakhand's population and contributes about 8.45 per cent of State GDP, yet occupies only 13 per cent of the land area, some 6.96 lakh hectares, of which just 3.45 lakh hectares are irrigated and 85.83 per cent of that irrigation lies in the plains.

The consequence is a sharply divided sector: market-oriented irrigated systems in the Tarai and Bhabar, and rain-fed, fragmented, spring-dependent hill systems where more than 55 per cent of net sown area receives no assured irrigation and where climate stress, out-migration and declining returns are eroding viability together.

Impact illustration



Agriculture — Observed Change and Impact Illustration



Impact pathway



Foodgrain production has remained broadly stable at 1.61-1.98 million tonnes and productivity has risen 37 per cent since 2011-12, but the composition of that resilience is shifting: high-value horticulture has expanded (fruit productivity from about 5 to 9 t/ha) while traditional hill foodgrain area has contracted by more than a quarter. The vulnerability is therefore concentrated not in aggregate output but in the rain-fed hill systems and orchard belts where climate stress, out-migration and declining returns interact.

Climate hazard	Climate driver	Climate stressor	Direct impact	Functional impact	Consequences	Confidence
Rising temperature and chilling deficit	Increasing mean and winter minimum temperatures	Accumulated winter chilling falls below cultivar requirement;	Delayed, protracted and irregular bud break; poor flower quality	Loss of commercial viability of established cultivars below a	Apple production -47% (2013-2023); apple area -55%; near-doubling of establishment cost at	High

	with elevation-dependent amplification (+1.5°C since 1970)	accelerated and desynchronised phenology	and low fruit set in apple, pear, walnut, apricot, plum and cherry	threshold altitude; upward migration of the viable orchard belt	higher altitude; stranded perennial capital	
Erratic and delayed monsoon	Shifting onset and cessation; longer intra-seasonal dry spells; ~12% decline in total monsoon rainfall since 2000	Soil-moisture deficit at germination and tillering; reduction in infiltration-dominant rainfall	Germination failure; reduced tillering and grain-fill in rain-fed millets, maize, wheat and paddy	Rising inter-annual yield variance on the >55% of cultivated area without irrigation access	Mandua –27%, jhangora –20%, barley –11% (2012–13 to 2021–22); income volatility; erosion of subsistence food security	High
Extreme rainfall and cloudburst	~35% increase in events exceeding 100 mm/day; cloudburst incidence 11 → 83 events	Runoff exceeding terrace infiltration capacity; slope saturation and mass movement	Terrace scour and riser failure; crop burial; destruction of guhl channels; landslide loss of cultivable land	Permanent reduction in net cultivable area; multi-season loss of soil productive capacity	44,882 ha destroyed in 2023 (~5.6% of cultivated area); multi-season recovery burden; abandonment of damaged terraces	High
Hailstorm	Increasing convective instability in the pre-monsoon and early monsoon period	Mechanical impact energy delivered during the fruit-development window	Injury to fruit skin, foliage and bearing shoots; destruction of marketable grade	Loss of the entire marketable consignment even where biological yield partially survives	75–100% crop loss in badly affected orchards from single events; acute and largely uninsured income shock	High
Drought and soil-moisture deficit	Rainfall deficit combined with higher evaporative demand under warming	Prolonged root-zone moisture deficit; declining surface-water reliability	30–40% yield reduction in drought years; pasture and fodder biomass loss	Loss of a full production season with no irrigation buffer in the hills; transfer of the deficit to the aquifer in the Tarai	10 of 13 districts affected 2007–09; Tarai groundwater draft +66% (0.99 → 1.64 BCM); 4–6 additional hours per day of water collection borne by women	High
Spring discharge decline and glacier retreat	Reduced recharge, declining snowfall fraction and sustained negative glacier mass balance	Falling lean-season discharge in springs, gadhera and glacier-fed canals; seasonal shift away from the rabi window	Contraction of the command area of gravity-fed gul and nali irrigation; ~12,000 springs dried statewide	Progressive reversion of assured-irrigated land to rain-fed status, compounding every rainfall-driven pathway above	Land abandonment where no alternative source exists; peak-water transition around 2050 in glacier-dependent commands in Chamoli, Uttarkashi, Pithoragarh and Tehri Garhwal	Medium–High
Warming winters and snow retreat	Rising winter minima; declining snow-covered area and duration	Loss of the insulating snowpack while radiative frost nights persist; reduced	Frost injury to orchards despite rising seasonal means; altitudinal expansion of red spider mite	Increased crop-protection cost and input dependence; yield and quality loss at high altitude	Rising cost of production at precisely the altitudes to which cultivation is being forced to migrate	Medium–High

		overwintering pest mortality	and other pests			
Forest fire and pasture degradation	Warmer, drier pre-monsoon conditions and an extended fire season	Loss of grazing biomass, lopped fodder and NTFP resource in the lean pre-monsoon period; post-fire ground-cover loss	Fodder deficit at the point of maximum seasonal scarcity; increased runoff and reduced slope cohesion above terraces	Reduced livestock productivity and distress sale; elevated terrace-failure probability in the monsoon of the same year	>1,100 major incidents in 2024; severe-fire-risk days projected to rise ~60%; compound fire-landslide loss sequence within a single year	High (fire); Medium (fodder link)
Heat and humidity stress on livestock and sericulture	Rising Temperature–Humidity Index in the Tarai–Bhabar belt and lower valleys; warmer autumn rearing season	THI exceeding the 72 and 78 thresholds; September–October rearing temperatures persistently above the 26°C optimum	Measurable milk-yield decline above THI 72; sharp productivity, conception-rate and welfare decline above THI 78; reduced cocoon shell weight and elevated larval mortality	Loss of the commercially critical autumn silk crop; reduced reliability of dairy income; phenological mismatch between mulberry leaf flush and rearing cycle	Erosion of the secondary livelihood on which over 70% of rural households depend; elevated muscardine and pebrine outbreak risk capable of causing complete batch loss	Medium–High
Climate-driven wildlife range shift	Altered forest fodder and fruiting phenology; reduced black-bear hibernation duration; forest-edge advance into abandoned land	Increased crop-raiding pressure from wild boar, rhesus macaque, nilgai, porcupine and bear; increased livestock predation	16–24% productivity loss for maize, wheat and paddy in conflict zones; livestock mortality from bear and leopard predation	12–17% reduction in net cropped area relative to non-conflict areas; diversion of household labour to crop guarding	Over 50% of erstwhile cultivated land lying barren in some Pauri Garhwal and Almora villages; guarding burden falling on women and the elderly; PMFBY revised in 2025 to insure the risk	Medium–High
Compound hazards (chronic decline + episodic destruction + labour scarcity)	Convergence of slow-onset productivity decline, episodic asset destruction and demographic depletion	Simultaneous loss of yield, of the productive asset and of the labour required to reconstruct it	Terraces damaged in one season are not reconstructed in the next; orchards removed are not replanted	Transition from recoverable loss to permanent abandonment, the sensitivity threshold beyond which the system does not return	1,792 ghost villages by 2022 (from 1,034 in 2011); unmaintained terraces raise landslide and erosion risk on slopes formerly regulated by active cultivation	High

Hazard, exposure, vulnerability and risk

Climate and Hazards

- Rising mean and maximum temperatures
- Erratic monsoon rainfall and prolonged dry spells
- High-intensity rainfall and flash floods
- Landslides and debris flows

- Riverine flooding in foothill and Tarai blocks
- Declining winter chill and changing snowfall patterns
- Hailstorms during flowering and fruiting seasons
- Increasing forest fire incidence
- Increased frequency of droughts

Exposures

- Rainfed agriculture farming systems
- Agricultural workers and smallholder farmers
- Horticulture
- Field crops
- Fisheries and aquaculture
- Dairy production systems
- Livestock and fodder resources
- Horticulture
- Beekeeping

Vulnerabilities

- High dependence on monsoon rainfall
- Small and fragmented landholdings
- Degraded and erosion-prone hill slopes
- Declining spring discharge
- Limited irrigation coverage in hill blocks
- High dependence on ecosystem services (pollination, forests, soil moisture)
- Climate-sensitive horticultural production
- Limited access to climate-resilient technologies and markets
- Limited crop diversification and climate-resilient seed access
- Limited adaptive capacity of marginal and smallholder farmers
- Financial Stability of Producers

RISKS

- Declining yields of rainfed crops
- Increasing livestock heat and cold stress
- Shifts in suitable cultivation zones and horticulture zones
- Reduced fruit quality and chilling-dependent crop yields
- Increased water competition and seasonal water stress
- Soil erosion
- Increased Wild animal attacks
- Damage to crops, irrigation systems and farm infrastructure
- Reduced livelihood security
- Out-migration from climate-stressed farming communities

Adaptation observations and considerations

- Treat the orchard belt transition as a planned adaptation, not an incidental adjustment. Chilling-hour deficits are already displacing apple cultivation upslope in Almora, Chamoli, Nainital, Bageshwar and Pithoragarh; low-chill

and disease-resistant cultivar programmes under the Kiwi Policy need to be targeted using block-level vulnerability results rather than district averages.

- Prioritise moisture security over irrigation expansion in the hills. Steep slopes, shallow soils and fragmented holdings constrain conventional irrigation; springshed recharge, micro-irrigation, water harvesting and soil-moisture conservation deliver more per rupee in mid-hill blocks than canal extension.
- Use the Millet Mission as a climate instrument, not only a nutrition one. Traditional cereals: mandua, jhangora, amaranth are intrinsically better matched to erratic rainfall than the crops displacing them; procurement and market linkage matter more than promotion alone in reversing the 27 per cent decline in traditional foodgrain area.
- Link crop insurance triggers to the hazards that actually cause loss. Hailstorm, frost, cloudburst and dry spell are the dominant loss mechanisms in hill agriculture; parametric triggers calibrated to these, with fast-tracked settlement, would improve on area-yield approaches designed for plains conditions.
- Address farm abandonment as a driver of secondary risk. Terrace collapse, loss of soil conservation labour and intensifying human-wildlife conflict follow from abandonment; convergence between MGNREGA-financed terrace repair, USRLM livelihood support and corridor-based conflict mitigation addresses the cause rather than the symptom.

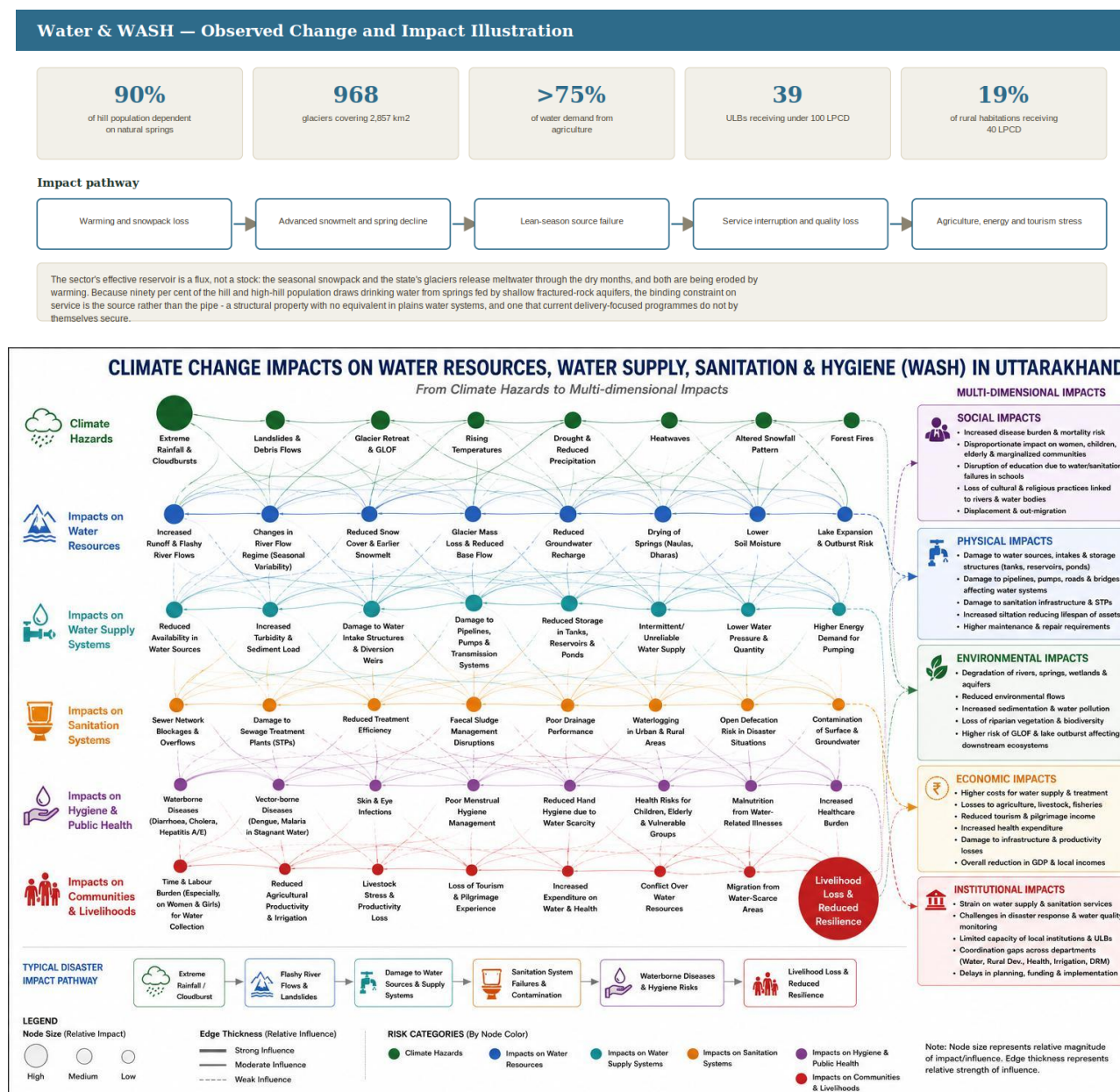
5. Water Resources & WASH

Key points

Uttarakhand holds 968 glaciers across 2,857 km², 17 major rivers, 31 lakes and 8 catchments comprising 116 sub-watersheds and 1,110 micro-watersheds, an endowment that is overwhelmingly a seasonal flux rather than a stored volume, and one that warming is steadily eroding.

Ninety per cent of the hill and high-hill population draws drinking water from natural springs: dhara, naula, umre ko pani- fed by shallow fractured-rock aquifers. The binding constraint on service is therefore the source, not the pipe, a structural property with no equivalent in plains water systems.

Impact illustration



Climate hazard	Climate driver	Climate stressor	Direct impact	Functional impact	Consequences	Confidence
Spring discharge decline	Rising temperatures, changing	Reduced aquifer recharge and	Reduced source yield;	Intermittent or inadequate household water	Longer water-collection	High

and groundwater recharge stress	precipitation, declining snowfall and reduced recharge in spring-shed catchments	falling lean-season discharge from springs, dharas and small streams	seasonal drying of rural drinking-water sources	supply; increased dependence on alternative and tanker sources	time, higher O&M costs, pressure on household water security; disproportionate burden on women and girls	
Erratic and delayed monsoon	Shifting monsoon onset/withdrawal and longer intra-seasonal dry spells	Longer periods between recharge events and greater variability in source flows	Reduced reliability of spring, surface-water and shallow groundwater sources	Water-supply schemes designed around historical source yields experience seasonal shortages	Reduced service-level reliability; increased need for storage, source augmentation and emergency supply	High
Drought and prolonged dry spells	Rainfall deficit combined with higher evaporative demand under warming	Persistent soil-moisture and groundwater deficit; reduced baseflow	Drying of springs, streams and shallow wells; reduced raw-water availability	Reduced hours/days of piped supply and increased dependence on stored water	Household water insecurity, hygiene compromise, increased collection burden and higher expenditure on alternative water	High
Extreme rainfall and cloudburst	Increasing intensity/frequency of short-duration heavy rainfall events	Runoff exceeds drainage and infiltration capacity; rapid sediment and debris transport	Damage to intake structures, pipelines, reservoirs, treatment units and distribution networks	Temporary or prolonged interruption of drinking-water services	Emergency repair costs, loss of safe water access and prolonged service disruption in affected villages and towns	High
Flooding and flash floods	More intense precipitation and rapid runoff in steep	Floodwater inundation, erosion and debris	Contamination or physical destruction	Loss of potable water supply and sanitation functionality	Increased exposure to faecal contamination	High

	Himalayan catchments	deposition around water and sanitation infrastructure	of water sources, wells, pipelines, toilets and sewerage infrastructure	during and after floods	on and water-borne disease; emergency WASH requirements in disaster-affected communities	
Landslides and slope instability	Extreme rainfall, rainfall concentration, freeze–thaw changes and loss of slope stability	Ground movement, debris flow and road/slope failure affecting linear infrastructure	Breaking, burying or exposing transmission and distribution pipelines; damage to access roads and treatment facilities	Isolation of water-supply schemes and difficulty accessing sites for repair and chlorination	Long repair periods, high infrastructure replacement costs and repeated service interruptions in mountain settlements	High
Glacier retreat, snowpack decline and cryospheric change	Rising temperatures, declining snow accumulation and sustained glacier mass loss	Altered seasonal runoff and changes in glacier/snow melt contribution to downstream flows	Changes in source-water availability and seasonal flow timing in glacier- and snow-fed systems	Greater uncertainty in long-term source reliability, particularly for high-altitude and downstream systems	Need for diversified water sources, additional storage and climate-resilient water-supply planning; potential long-term water-security stress	Medium–High
Climate-driven water-quality deterioration	Higher temperatures, intense rainfall, flooding, erosion and prolonged low-flow conditions	Increased sediment, turbidity, microbial contamination and concentration of pollutants during low-flow periods	Deterioration of raw-water quality and increased treatment requirements	Treatment plants and community systems face higher treatment loads; untreated sources become unsafe	Increased water-treatment costs, greater risk of diarrhoeal and other water-borne diseases, and reduced confidence	High

					in drinking-water safety	
Warming and heat stress	Rising mean and extreme temperatures, particularly in lower valleys and the Tarai-Bhabar belt	Higher household water demand and increased water requirements for hygiene, cooling and institutional uses	Increased per-capita water demand and faster depletion of household storage	Existing systems experience higher peak demand and greater pressure on sources and distribution networks	Seasonal shortages, increased pumping/energy costs and reduced ability of vulnerable households to maintain adequate hygiene	Medium–High
Forest fire and catchment degradation	Warmer and drier pre-monsoon conditions and longer fire seasons	Loss of vegetation cover, increased erosion and reduced infiltration in recharge zones	Increased sedimentation and reduced recharge in spring-shed and catchment areas	Declining source reliability and higher turbidity following rainfall events	Progressive deterioration of drinking-water sources, increased treatment/O &M burden and reduced resilience of rural water systems	Medium–High
Sanitation-system disruption during extreme events	Extreme rainfall, flooding, landslides and prolonged water stress	Flooding/structural damage to toilets, septic tanks, drains and sewerage networks; inadequate water for flushing and cleaning	Overflow, leakage or failure of sanitation infrastructure	Reduced containment of human excreta and disruption of wastewater management	Increased faecal contamination of surface and groundwater, environmental pollution and public-health risk	High
Water scarcity and hygiene stress	Declining source reliability combined with higher temperatures and rainfall variability	Reduced quantity of water available for bathing, handwashing, cleaning and menstrual hygiene	Households prioritise drinking and cooking over non-potable uses	Reduced frequency of handwashing, bathing and cleaning of toilets and domestic environments	Increased hygiene vulnerability, particularly for women, children, older people and other water-dependent groups	High

Compound WASH hazards: drought + extreme rainfall + infrastructure fragility	Convergence of slow-onset drying, episodic heavy rainfall, landslides and ageing/fragile infrastructure	Simultaneous loss of source quantity, deterioration of water quality and physical damage to supply systems	Water sources become both less available and less reliable/safe	Communities move repeatedly between water scarcity and emergency contamination/service disruption	Transition from recoverable seasonal disruption to chronic WASH insecurity; rising adaptation and O&M costs and increased inequality between accessible and remote settlements	High
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Hazard, exposure, vulnerability and risk

Climate and Hazards

- Rising temperatures and increasing evapotranspiration
- Changing monsoon patterns and rainfall variability
- Increased drought frequency
- High-intensity rainfall and cloudburst events
- Landslides affecting water sources and infrastructure
- Glacier retreat and declining snowmelt contribution
- Drying of springs and reduced groundwater recharge

Exposures

- Rural and urban communities dependent on springs and groundwater
- Drinking water supply schemes and gravity-fed distribution systems
- Reservoirs and groundwater sources
- Piped water distribution networks
- Household drinking water connections
- Sanitation facilities and wastewater infrastructure
- Schools, health facilities and community WASH services
- Watersheds, recharge zones and freshwater ecosystems
- Irrigation and multi-purpose water infrastructure
- Water supply and sanitation infrastructure located in flood- and landslide-prone areas

Vulnerabilities

- High dependence on climate-sensitive seasonal sources
- Seasonal water scarcity and declining source reliability
- Degraded catchments reducing recharge
- Increasing competition among domestic, agricultural, tourism and ecological water demands
- Limited storage, treatment and distribution capacity in hill settlements

- Limited real-time monitoring of water quantity and quality
- High dependence on monsoon recharge

RISKS

- Drying of springs and declining groundwater availability
- Frequent disruption of piped water supply due to damaged pipelines and intake structures
- Damage to water supply, sanitation and drainage infrastructure
- Increased incidence of water-borne and sanitation-related diseases following floods and intense rainfall
- Reduced reliability of WASH services in schools, health facilities and communities
- Forest degradation and soil erosion reducing watershed recharge capacity
- Reduced resilience of communities, ecosystems and livelihoods to climate extremes

Adaptation observations and considerations

- Close the governance gap between delivery and source. Jal Jeevan Mission secures the distribution infrastructure while the hydrological source on which it depends is protected by a separate and weaker instrument; springshed security needs to become a funded, accountable component of scheme design rather than a nominal coordination requirement.
- Scale springshed management using the NITI Aayog protocol, targeted by vulnerability. Recharge-area treatment, Van Panchayat catchment management and MGNREGA-financed watershed works should be sequenced onto the blocks where spring dependence and discharge decline coincide, not distributed uniformly.
- Plan for a shift in the timing, not only the volume, of supply. Advancing snowmelt means runoff increasingly arrives before the pre-monsoon and summer peak in agricultural, domestic and pilgrimage demand; storage, transfer capacity and demand management need to be assessed against seasonal misalignment rather than annual totals.
- Climate-proof the intake, not just the network. Gravity-fed schemes fail at the source and the headworks under cloudburst and landslide conditions; flood-resilient intake design, slope protection above collection chambers and redundant alternate sources deserve priority in high-hazard blocks.
- Treat sanitation reversal as a climate risk. Post-event damage to household latrines and water quality contamination in flood-affected habitations threatens ODF status and drives waterborne disease; seismic and slope-rated retrofit standards, and post-event water quality surveillance, close a gap that current programmes do not cover.

6. Roads & Linear Township

Key points

The PWD-maintained network totals 48,266.6 km, including 3,166 km of National Highways, 5,810 km of State Highways, 29,176 km of village roads and 2,954 km of border roads, with 13,916 villages now connected and the 736.994 km Char Dham All-Weather Road sanctioned at ₹10,431 crore.

Because the state has almost no navigable waterways and only nascent rail penetration into the hills, this dependence has no practical substitute: in the words of the SAPCC transport sub-strategy, any disruption to the road network translates directly into a humanitarian and economic emergency.

Impact illustration

Roads & Linear Township — Observed Change and Impact Illustration



Impact pathway



The PWD-maintained network has more than tripled since state formation, from 8,080 km of metalled road in 2000 to 26,701 km in 2025. That expansion has delivered access to 13,916 villages, but it has also placed a long, structurally exposed asset base on terrain that is 86 per cent mountainous and seismically active. Because alternative modes are largely absent, disruption has no practical substitute, and an ageing bridge stock compounds the consequence of any single corridor failure.

Climate Hazard	Climate Driver	Climate Stressor	Direct Impact	Functional Impact	Consequences	Confidence
Rising Temperature	Increasing average and summer temperatures and more frequent heat extremes	Thermal expansion, pavement softening and accelerated material deterioration	Rutting, cracking and reduced service life of flexible pavements; thermal stress on bridges and road structures	Increased maintenance requirements and reduced structural performance of transport infrastructure	Higher maintenance costs, reduced asset lifespan and disruption to transport services	Medium–High
Changing Rainfall Pattern	Erratic monsoon onset and withdrawal and uneven rainfall distribution	Reduced drainage efficiency, fluctuating soil moisture and prolonged wet conditions	Accelerated pavement deterioration, shoulder erosion and drainage failure	Reduced road serviceability and increased maintenance frequency	Increased travel delays, higher maintenance expenditure and reduced transport reliability	Medium–High
Extreme Rainfall / Cloudbursts	High-intensity precipitation events and cloudbursts	Surface runoff exceeding drainage capacity, debris flow and slope failures	Washout of roads, culverts and retaining structures; bridge scour and embankment erosion	Closure of highways, district roads and rural roads and interruption of traffic movement	Transport disruption, economic losses and delayed emergency response	High
Flooding / Flash Floods	River overflow and extreme runoff	Inundation of roads, bridges and low-lying	Damage to bridges, culverts, road embankments	Loss of transport connectivity and	Isolation of communities, disruption of supply chains	High

		transport corridors	and river-crossing infrastructure	prolonged restoration of road services	and increased reconstruction costs	
Landslides & Slope Instability	Intense or prolonged rainfall and unstable mountain slopes	Rockfalls, debris slides and slope failures along road corridors	Damage to carriageways, retaining walls, tunnels and protective structures	Reduced accessibility and prolonged closure of strategic transport corridors (e.g., Char Dham routes)	Isolation of settlements, disruption of tourism and trade and increased maintenance burden	High
Glacier Retreat & Permafrost Degradation	Rising temperatures, declining glacier mass and changing freeze–thaw cycles	Increased slope instability, glacial sediment movement and changing hydrological regimes	Damage to high-altitude roads, bridges and protective infrastructure	Reduced reliability of transport infrastructure in glacier-fed catchments	Increased maintenance costs, higher infrastructure risk and reduced connectivity to remote regions	Medium
River Morphological Change & Sedimentation	Intense rainfall, erosion and changing river dynamics	Channel migration, aggradation and increased bridge scour	Undermining of bridge foundations, culvert blockage and embankment instability	Reduced structural safety of transport infrastructure	Increased bridge failures, traffic restrictions and expensive rehabilitation works	Medium–High
Forest Fires	Warmer, drier pre-monsoon conditions	Fire damage, smoke and reduced visibility along road corridors	Damage to roadside safety infrastructure, power supply to transport systems and reduced driver visibility	Increased accident risk and temporary road closures	Disruption of transport operations and increased emergency management costs	Medium
Freeze–Thaw Variability	Reduced winter snow cover and fluctuating temperatures around freezing point	Repeated freeze–thaw weathering of pavement and slope materials at higher elevations	Pavement cracking, pothole formation and rockfall initiation	Reduced pavement quality and increased slope maintenance requirements	Higher maintenance costs and increased road safety risks	Medium
Compound Climate Hazards (Extreme Rainfall + Floods + Landslides)	Increasing frequency and convergence of multiple climate extremes	Simultaneous failure of slopes, bridges, drainage systems and transport infrastructure	Widespread damage across road networks and prolonged closure of major transport corridors	System-wide disruption of passenger movement, freight transport and emergency access	Long-term connectivity loss, severe economic impacts, disruption of tourism and reduced transport system resilience	High



Hazard, exposure, vulnerability and risk

Climate and Hazards

- Rising temperatures and increasing heat stress
- High-intensity rainfall and cloudburst events
- Flash floods and riverine flooding
- Landslides, rockfalls and slope failures
- Debris flows and riverbank erosion
- Glacial Lake Outburst Floods (GLOFs) in upper catchments
- Freeze-thaw cycles and changing snowfall patterns
- Increasing forest fire incidence affecting transport corridors

Exposures

- National Highways, State Highways and Major District Roads
- Rural roads and PMGSY connectivity
- Bridges, tunnels, culverts and retaining structures
- Roadside settlements and linear townships
- Critical lifeline infrastructure (power, telecom, water supply and pipelines) along road corridors
- Public transport and emergency evacuation routes
- Tourism, pilgrimage and freight transport corridors
- Schools, hospitals and public services dependent on road connectivity

Vulnerabilities

- Road corridors traversing steep, unstable and erosion-prone terrain
- Linear settlements concentrated along river valleys and highways
- High dependence on single-access road corridors in remote blocks
- Ageing bridges, culverts and inadequate drainage infrastructure
- Unplanned roadside development and slope modifications

- Limited alternate routes and network redundancy
- Inadequate slope stabilization and bioengineering measures
- Increasing traffic loads exceeding infrastructure design capacity
- Difficult terrain delaying maintenance, emergency response and restoration
- Limited climate-resilient planning, monitoring and asset management

IMPACTS AND RISKS

- Frequent road closures and prolonged disruption of connectivity
- Damage to roads, bridges, tunnels and retaining structures
- Isolation of remote communities and disruption of essential services
- Disruption of tourism, pilgrimage, trade and local economies
- Increased travel time, transport costs and freight delays
- Damage to power, water supply and telecommunication networks along transport corridors
- Reduced accessibility to healthcare, education and emergency response
- Increased accident risk from landslides, rockfalls and flash floods
- Increased maintenance and reconstruction costs
- Reduced resilience of linear townships and critical transport networks

Adaptation observations and considerations

- Recognise construction as a hazard driver, not only a hazard victim. Evidence from the sector shows cumulative slope destabilisation from distributed construction and a feedback between road investment and geotechnical failure; slope-stability appraisal and bioengineering need to be embedded in alignment and design decisions rather than applied as post-failure remediation.
- Size drainage to intensified rainfall, not historical rainfall. Culvert and cross-drainage capacity based on past intensity-duration statistics is the most common point of failure under cloudburst conditions; retrofit programmes should be prioritised on corridors combining high extreme-rainfall hazard with high landslide susceptibility.
- Build redundancy where a single corridor carries the whole consequence. Blocks and villages with single, non-redundant road access convert a local failure into a total access failure; identifying and progressively eliminating single-corridor dependence is a higher-value investment than incremental widening.
- Manage the ageing bridge stock as a distinct risk class. About 818 inventoried bridges — 23.5 per cent of the total — are already 40 years or older, and Bailey bridge dependence indicates unresolved permanent-structure gaps; a condition-based, hazard-weighted replacement programme is required.
- Regulate ribbon development along road-influence zones. Road expansion has drawn settlement onto marginal, high-susceptibility terrain, concentrating exposure precisely where slopes are least stable; planning regulation in road-influence zones addresses the exposure that engineering alone cannot.

7. Health

Key points

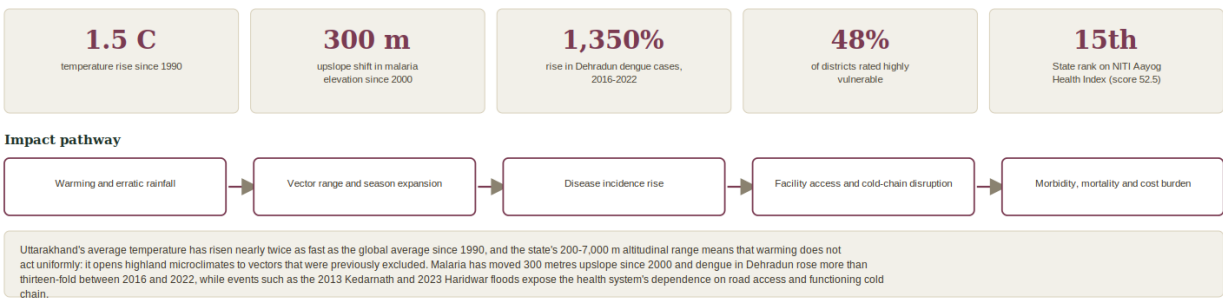
Uttarakhand's average temperature has risen nearly twice as fast as the global average since 1990, and its altitudinal range from 200 to 7,000 metres creates microclimates that respond very differently to that warming — opening highland areas to disease vectors that were previously excluded.

The consequences are already measurable: malaria has shifted 300 metres upslope since 2000, dengue cases in Dehradun rose 1,350 per cent between 2016 and 2022, diarrhoeal disease rose about 30 per cent during monsoon months in Uttarkashi and Chamoli, and 48 per cent of districts are rated highly vulnerable.

Lead department(s): Department of Health & Family Welfare; State Health Mission; National Health Mission

Impact illustration

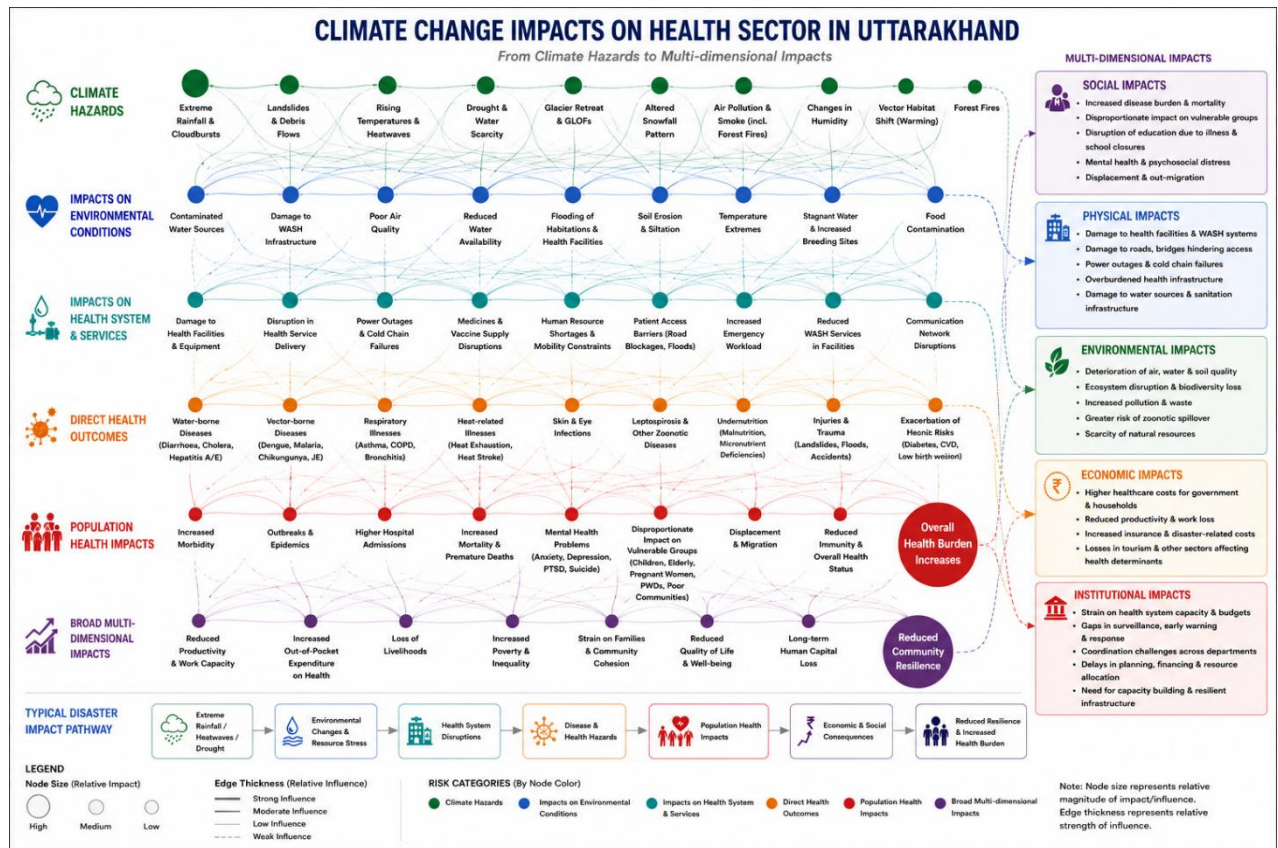
Health — Observed Change and Impact Illustration



Climate Hazard	Climate Driver	Climate Stressor	Direct Impact	Functional Impact	Consequences	Confidence
Rising Temperature	Increasing average and maximum temperatures and more frequent heatwaves	Heat stress, warmer nights and prolonged exposure to high temperatures	Increased cases of heat-related illnesses, dehydration and exacerbation of cardiovascular and respiratory diseases	Increased demand for emergency care, outpatient services and hospital admissions	Higher heat-related morbidity and mortality, particularly among elderly, children, outdoor workers and vulnerable populations	High
Changing Rainfall Pattern	Erratic monsoon onset and withdrawal and uneven rainfall distribution	Seasonal water scarcity, declining water quality and sanitation challenges	Increased incidence of diarrhoeal diseases, dehydration and nutritional stress	Increased burden on primary healthcare facilities and public health programmes	Higher disease burden, especially among children and rural communities	High
Extreme Rainfall / Cloudbursts	High-intensity precipitation events and cloudbursts	Flooding of settlements, contamination of drinking water sources and localized infrastructure damage	Outbreaks of water-borne diseases and injuries associated with extreme weather events	Increased emergency response requirements and surge in healthcare demand	Increased morbidity, localized mortality and pressure on district health systems	High

Flooding / Flash Floods	River overflow and extreme runoff	Damage to health facilities, roads, utilities and communication networks	Disruption of hospitals, Primary Health Centres (PHCs), ambulance services and medical supply chains	Reduced accessibility and continuity of essential health services	Delayed treatment, increased mortality risk and interruption of maternal, child and emergency healthcare services	High
Landslides & Slope Instability	Intense or prolonged rainfall and unstable mountain slopes	Physical isolation of remote settlements and damage to transport infrastructure	Increased trauma cases and disruption of referral transport systems	Reduced access to healthcare facilities and delayed emergency medical response	Increased preventable deaths, prolonged recovery and widening health inequities in remote mountain communities	High
Vector-borne Disease Expansion	Rising temperatures and changing rainfall regimes	Expansion of suitable habitats and longer breeding seasons for disease vectors	Increased transmission of dengue, malaria, scrub typhus and other vector-borne diseases into higher elevations	Increased disease surveillance, diagnostics and vector-control requirements	Expansion of disease burden into previously unaffected areas and increased public health expenditure	High
Forest Fires	Warmer, drier pre-monsoon conditions	Smoke, particulate matter (PM _{2.5}) and reduced air quality	Increased respiratory illness, asthma exacerbation and cardiovascular complications	Increased hospital admissions and respiratory healthcare demand	Higher respiratory morbidity among vulnerable populations and increased healthcare costs	Medium–High
Cold Waves & Temperature Extremes	Greater seasonal temperature variability and cold spells at higher elevations	Cold stress and prolonged exposure to low temperatures	Increased hypothermia and worsening of cardiovascular and respiratory illnesses	Increased seasonal healthcare demand and pressure on health services	Increased winter morbidity and mortality among elderly and vulnerable populations	Medium
Climate-induced Ecosystem Change	Altered temperature and rainfall regimes affecting ecosystems and wildlife	Changing distribution of disease vectors, reservoirs and zoonotic pathogens	Increased risk of zoonotic disease emergence and altered infectious disease patterns	Enhanced surveillance, laboratory capacity and One Health coordination requirements	Increased public health risk and greater complexity of disease management	Medium

Compound Climate Hazards (Heat + Extreme Rainfall + Floods + Landslides)	Increasing frequency and convergence of multiple climate extremes	Simultaneous disruption of healthcare infrastructure, workforce, supply chains and emergency response	Reduced continuity of essential health services during disaster events	System-wide stress on the health system and reduced emergency response capacity	Increased mortality, prolonged service disruption, mental health impacts and reduced health system resilience	High
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Hazard, exposure, vulnerability and risk

Climate and Hazards

- Rising mean and maximum temperatures
- Increasing frequency and intensity of heatwaves
- Erratic monsoon rainfall and prolonged dry spells
- High-intensity rainfall, cloudbursts and flash floods
- Riverine flooding in foothill and Tarai regions
- Landslides disrupting transport and healthcare access
- Increasing forest fire incidence and deteriorating air quality
- Changing distribution of vector-, water- and food-borne diseases

Exposures

- Rural and urban populations
- Children, older adults, pregnant women and persons with disabilities
- Outdoor workers and climate-sensitive livelihoods
- Hospitals, Community Health Centres (CHCs), Primary Health Centres (PHCs) and Sub-Centres

- Ambulance services and emergency referral networks
- Health supply chains, medicines and cold-chain infrastructure
- Drinking water, sanitation and hygiene (WASH) systems supporting health services
- Remote mountain communities, pilgrimage routes and seasonal tourist populations

Vulnerabilities

- Remote and difficult-to-access settlements with limited healthcare accessibility
- Health facilities located in flood-, landslide- and erosion-prone areas
- Limited emergency preparedness and surge capacity during extreme events
- Inadequate climate-resilient health infrastructure, backup power and water supply
- Unequal access to quality healthcare across hill and plain regions
- High burden of climate-sensitive diseases and vulnerable population groups
- Shortage of healthcare professionals in remote blocks
- Disruptions to transport and medical supply chains during disasters
- Limited disease surveillance, early warning and climate-informed health planning
- Socio-economic inequalities affecting health outcomes and adaptive capacity

IMPACTS AND RISKS

- Increased heat-related illnesses and mortality
- Expansion of vector-, water- and food-borne diseases
- Disruption of healthcare services during floods, landslides and extreme weather
- Damage to hospitals, PHCs, CHCs and health infrastructure
- Reduced accessibility to emergency, maternal and routine healthcare services
- Interruptions in essential medicines, vaccines and emergency medical supplies
- Increased demand for emergency response and public health services
- Deterioration of maternal, child and elderly health outcomes
- Increased mental health and psychosocial impacts following disasters
- Increased public health expenditure and reduced resilience of the healthcare system

Adaptation observations and considerations

- Extend surveillance to the altitudes that are newly at risk. Vector and entomological surveillance designed around historical transmission zones no longer matches the distribution of risk; upslope and border blocks such as Dharchula and Munsyari combine rising exposure with the thinnest surveillance coverage.
- Treat facility access, not only facility count, as the binding constraint. A share of the population lies beyond 5 km of a health facility, and landslide or flood closure of a single corridor can isolate a block entirely; redundant emergency transport, helipad access and telemedicine linkage address the failure mode that additional beds alone do not.
- Protect the cold chain against disruption, not just against outage. Vaccine and medicine integrity depends on power and access simultaneously; solar and portable cold-chain capacity at remote PHCs is a low-cost measure with disproportionate benefit during the monsoon peak.
- Bring post-disaster mental health into routine service. Post-event psychological morbidity is documented in the state yet remains among the least-resourced and least-measured impact pathways; community-based models, including AYUSH-linked approaches, offer a scalable route beyond the concentration of psychiatric services in Dehradun and Nainital.
- Align health planning with the SAPCCHH and with block-level risk results. Staffing norms, CHC:PHC ratios against the IPHS 1:4 benchmark, and doctor and bed density should be assessed against block vulnerability rather than population alone, so that scarce human resources are deployed where compound risk is highest.

8. Energy

Key points

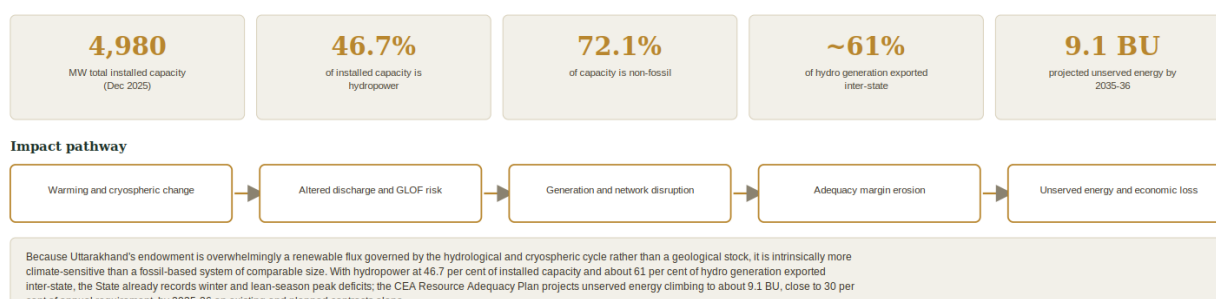
Installed capacity stands at about 4,980 MW, of which 72.1 per cent is non-fossil and hydropower alone accounts for 2,323.89 MW, or 46.7 per cent. Exploitable large-hydro potential is assessed at about 13,481 MW across 55 projects, with the first 500 MW of the 1,000 MW Tehri pumped-storage plant commissioned in 2025.

Because the state's endowment is overwhelmingly a renewable flux governed by the hydrological and cryospheric cycle rather than a geological stock, it is intrinsically more climate-sensitive than a fossil-based system of comparable size — and roughly 61 per cent of hydro generation is exported even as the state records winter and lean-season peak deficits.

Lead department(s): Uttarakhand Power Corporation Limited; Power Transmission Corporation of Uttarakhand Limited; Uttarakhand Jal Vidyut Nigam

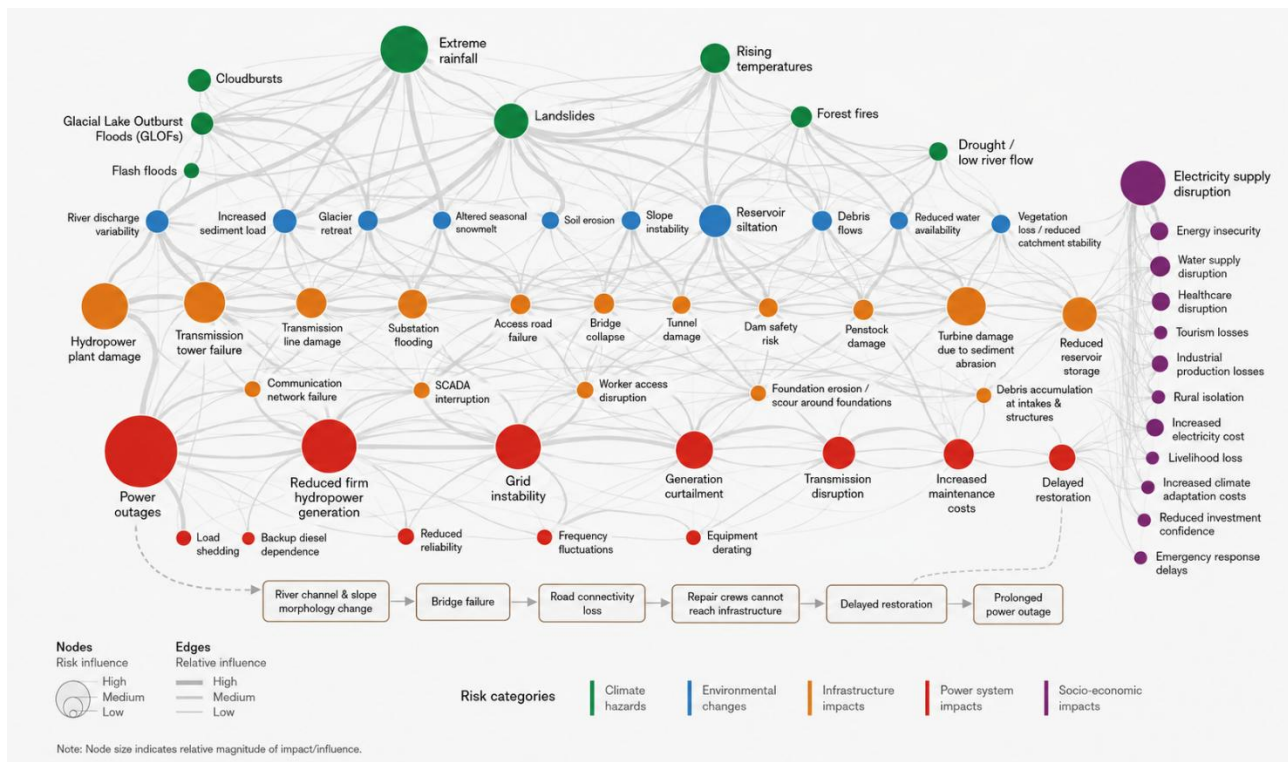
Impact illustration

Energy — Observed Change and Impact Illustration



Climate Hazard	Climate Driver	Climate Stressor	Direct Impact	Functional Impact	Consequences	Confidence
Rising Temperature	Increasing average and summer temperatures	Higher cooling demand and thermal stress on electrical equipment	Reduced efficiency of transformers, substations and transmission equipment	Increased peak electricity demand and reduced system efficiency	Higher operational costs, increased outage risk and pressure on grid reliability	High
Changing Rainfall Pattern	Erratic monsoon onset, prolonged dry spells and uneven rainfall distribution	Altered river discharge and seasonal water availability	Reduced dependable generation from run-of-river hydropower projects during lean seasons	Lower reliability of electricity generation and seasonal supply variability	Reduced energy security and increased dependence on external power sources	High
Drought / Reduced Lean-season Flows	Rainfall deficit, reduced snow accumulation and declining baseflows	Reduced river discharge available for hydropower generation	Declining hydropower output and turbine operating efficiency	Reduced availability of renewable electricity during peak demand periods	Power shortages, higher generation costs and reduced energy resilience	High

Extreme Rainfall	High-intensity precipitation and cloudburst events	High sediment load, debris flow and sudden inflow surges	Damage to hydropower intakes, desilting chambers and powerhouse infrastructure	Temporary shutdown of hydropower facilities and reduced generation capacity	Service interruptions, infrastructure repair costs and generation losses	High
Landslides & Slope Instability	Intense rainfall and unstable mountain slopes	Damage to transmission towers, distribution lines and access roads	Failure of transmission corridors and restricted maintenance access	Reduced reliability of power transmission and delayed restoration	Prolonged outages in remote communities and increased maintenance costs	High
Glacier Retreat & Changing Snowmelt	Rising temperatures and declining glacier mass	Altered seasonal river flow and increasing hydrological variability	Reduced long-term hydropower potential and unpredictable inflows	Reduced reliability of hydropower planning and reservoir operations	Long-term decline in hydropower generation potential and investment uncertainty	Medium–High
High Sediment Load	Glacier melt, erosion and intense rainfall	Increased sediment transport in rivers	Turbine abrasion, accelerated equipment wear and reduced plant efficiency	Increased maintenance frequency and forced shutdowns	Higher operating costs and reduced asset lifespan	High
Forest Fires	Warmer, drier pre-monsoon conditions	Fire damage to transmission and distribution infra	Damage to poles, conductors and communication systems	Disruption of electricity transmission and field operations	Localized power outages and increased restoration costs	Medium–High
Compound Climate Hazards (Heat + Extreme Rainfall + Landslides)	Increasing frequency and convergence of multiple climate extremes	Simultaneous disruption of hydropower generation, transmission and distribution systems	Cascading failures across the electricity network and emergency response constraints	Reduced reliability and resilience of the power system	Widespread outages, economic disruption and reduced resilience of critical services	High



Hazard, exposure, vulnerability and risk

Climate and Hazards

- Rising mean and maximum temperatures
- Erratic monsoon rainfall and changing river flow regimes
- High-intensity rainfall and cloudburst events
- Flash floods and riverine flooding
- Landslides, slope failures and debris flows
- Glacial Lake Outburst Floods (GLOFs) in upper catchments
- Declining snowfall and glacier retreat affecting seasonal runoff
- Increasing forest fire incidence affecting transmission corridors
- Strong winds and thunderstorms causing power disruptions

Exposures

- Hydropower generation facilities and reservoirs
- Dams, barrages and water conveyance systems
- Transmission lines, substations and distribution networks
- Rural and urban electricity consumers
- Critical facilities dependent on uninterrupted power (hospitals, water supply, telecommunications and emergency services)
- Industrial, commercial and tourism energy demand centres
- Renewable energy installations (solar, micro-hydro and distributed systems)
- Remote communities dependent on single-grid electricity supply

Vulnerabilities

- High dependence on hydropower and climate-sensitive river flows
- Energy infrastructure located in flood-, landslide- and GLOF-prone valleys
- Transmission lines traversing steep, unstable and forested terrain
- Limited redundancy in power transmission and distribution networks
- Ageing infrastructure and increasing maintenance requirements
- Sedimentation affecting hydropower reservoirs and intakes
- Remote and difficult-to-access assets delaying inspection and restoration
- Growing seasonal electricity demand associated with tourism and urbanisation
- Limited climate-resilient asset planning and condition monitoring
- Dependence on a limited number of major generation and transmission corridors

IMPACTS AND RISKS

- Reduced hydropower generation due to altered river flows and seasonal variability
- Damage to dams, intakes, substations, transmission lines and distribution infrastructure
- Frequent power outages and disruption of electricity supply
- Isolation of remote communities due to failure of energy infrastructure
- Increased sedimentation reducing hydropower efficiency and reservoir capacity
- Disruption of water supply, healthcare, communications and other critical services dependent on electricity
- Increased operation, maintenance and infrastructure restoration costs
- Reduced reliability and resilience of the state power system
- Economic losses to industries, tourism and local businesses due to power interruptions
- Increased energy security risks and dependence on alternative power sources during extreme events

Adaptation observations and considerations

- Treat the adequacy margin as a climate variable. The CEA Resource Adequacy Plan projects unserved energy rising from about 2.4 BU in 2025-26 to about 9.1 BU by 2035-36, close to 30 per cent of annual requirement on existing and planned contracts; capacity planning needs to price in climate-driven variability in hydro availability rather than assume historical flow.
- Build GLOF and cascading-failure protocols into asset operation. The 2021 Chamoli rock-ice avalanche demonstrated how rapidly an upstream cryospheric event disables downstream generation; automatic shutdown triggers, intake isolation and instrumented early warning on flagged glacial lakes are proportionate to the demonstrated consequence.
- Plan restoration around access, not repair time. In hill terrain the time to restore a substation or transmission line is frequently governed by road access rather than by the repair itself; siting decisions, access-route redundancy and pre-positioned clearance capability should be assessed jointly with the road sector.
- Account for sediment as an operational, not incidental, cost. Rising suspended sediment concentration reduces turbine life and output and raises desilting cost; sediment-informed operating protocols and catchment treatment upstream of intakes protect generation assets at source.
- Diversify away from single-point cryospheric dependence. Distributed solar, small hydro and pumped storage reduce the concentration of risk in large glacier-fed assets, while the Tehri reservoir's genuine seasonal storage buffer illustrates the adaptation value of storage over run-of-river configuration.

9. Conclusion and Way Forward

This assessment set out to convert an existing climate hazard profile into decision-ready evidence at block level. It has produced that evidence for 95 blocks across five sectors, together with a composite index, block-wise risk profiles and a Digital Vulnerability Atlas. What follows is an honest account of what worked, what proved difficult, what was learned and where the opportunities now lie.

What worked

- A single, consistently applied risk framing across all sectors made results comparable, so that a block's standing in agriculture can be read alongside its standing in roads or health without reconciling different methods.
- Iterative departmental consultation and expert working group review functioned as genuine quality control rather than endorsement: several indicator choices, data sources and sector assumptions were revised in response, and the results are more defensible for it.
- Deriving sectoral indices before compositing them preserved diagnostic value. A composite score alone would have told departments where to worry; the sectoral decomposition tells them which sector is driving that concern in a given block.

Challenges

- Block-level data availability was the principal constraint. Several indicators specified in the framework exist reliably only at district level, or sit within utility and departmental systems not published at block resolution — asset registers, feeder-level loss data, facility staffing rosters and reliability indices among them.
- Data quality required active scrutiny rather than assumption. Implausible values, unit and scale inconsistencies between districts, and columns empty across all blocks were encountered and had to be excluded and logged rather than silently corrected, which reduces coverage but protects credibility.
- Hazard layers and vulnerability indicators do not always share the same spatial resolution. Where hazard information was available only at district scale, the asymmetry between block-level vulnerability and district-level hazard has been stated explicitly rather than concealed by presenting a single combined figure.
- Slow-onset hazards are poorly served by existing monitoring. Spring discharge, chilling-hour accumulation and sediment load are consequential and under-measured, which limits how precisely adaptation can be targeted for exactly the stresses that are hardest to reverse.

Learnings

- Aggregate performance can conceal concentrated vulnerability. Agricultural productivity rose 37 per cent while traditional hill foodgrain area fell 27 per cent; a state-level indicator alone would have missed the transition entirely.
- The binding constraint is often not the one that is funded. In water, the pipe is financed while the spring is not; in energy and health, restoration time is governed by road access rather than by repair or clinical capacity. Cross-sector diagnosis changes what should be invested in.
- Development investment can create exposure. Road expansion delivered connectivity to 13,916 villages and simultaneously drew settlement onto unstable terrain and destabilised cut slopes. Adaptation planning that does not examine this feedback will under-estimate future risk.
- Institutions are configured for sudden-onset events, while several of the most damaging hazards are chronic. Reframing slow-onset climate stress as a standing planning input rather than an emergency is a structural rather than technical shift, and among the most consequential.

Opportunities

- The Digital Vulnerability Atlas can become a standing decision-support layer rather than a one-time deliverable, provided ownership, update cycles and institutional responsibility are formally assigned within SECCCD and the line departments.
- Convergence, not new schemes, is the most immediate financing route. JJM, MGNREGA, CAMPA, PMFBY, PMGSY, NHM, the Apple Mission and the Millet Mission already reach the relevant blocks; block-level risk results allow existing allocations to be steered rather than supplemented.

- The identified convergent hotspots: high-altitude blocks in Uttarkashi, Chamoli, Rudrapur and Pithoragarh offer a defensible basis for a first-tranche, multi-sector investment package, which is both more efficient and easier to finance than dispersed sectoral action.
- Establishing baselines for the indicators used here enables progress in reducing vulnerability to be tracked over time, converting this assessment from a snapshot into the first observation in a series.
- The blocks where evidence is currently weakest are known and enumerable. A targeted primary data programme -spring discharge monitoring, utility asset geolocation, facility-level health rosters — would close the specific gaps that most constrain precision, at modest cost relative to the investment decisions it would inform.

Taken together, the assessment provides Uttarakhand with a transparent chain of evidence from climate projection through hazard, impact and vulnerability to block-level risk, and from that risk to named adaptation measures and accountable departments. Its value will be determined less by the scores it reports than by whether those scores are used to steer plans, budgets and sequencing decisions in the blocks where compound risk is highest.

10. Adaptation Stocktake

This section presents the current state of climate adaptation in Uttarakhand to inform the prioritisation of adaptation investments and the development of sector-specific adaptation pathways. The stocktake reviews existing policies, programmes, institutional arrangements and ongoing initiatives to identify strengths, gaps and opportunities for building long-term climate resilience across the state.

Key Analysis

Climate adaptation planning in Uttarakhand is progressing across state, district and local levels.

Uttarakhand has established a policy foundation for climate action through the State Action Plan on Climate Change (SAPCC), the State Climate Change Centre, disaster management institutions and sector-specific programmes. Recent efforts increasingly integrate climate adaptation into agriculture, water resources, forests, disaster risk reduction and infrastructure planning. However, implementation remains uneven across districts and sectors.

Adaptation implementation remains fragmented across sectors and departments.

Many adaptation measures are implemented through existing departmental programmes rather than through coordinated climate-resilience planning. Limited convergence among departments can reduce efficiency, particularly where climate risks span multiple sectors such as water, energy, roads, agriculture and public health.

Adaptation needs to become increasingly risk-informed and evidence-based.

While numerous programmes strengthen resilience, systematic use of climate projections, hazard modelling and vulnerability assessments in planning and investment decisions remains limited. Integrating downscaled climate projections and block-level vulnerability assessments into development planning can improve prioritisation of adaptation investments.

Hills-specific adaptation requires integrated landscape approaches.

Climate risks in Uttarakhand are closely linked across watersheds, forests, glaciers, rivers and settlements. Adaptation therefore requires coordinated planning across catchments and administrative boundaries to avoid transferring risk between upstream and downstream communities.

Strengthening critical infrastructure resilience is becoming increasingly important.

Increasing frequency of floods, landslides, glacial hazards, forest fires and extreme rainfall highlights the need to climate-proof roads, bridges, electricity networks, drinking water systems, health facilities and communication infrastructure through resilient design standards and proactive maintenance.

Community-based adaptation remains central to resilience.

Local institutions, Gram Panchayats, Van Panchayats, women's self-help groups, farmer producer organisations and community disaster management committees play a critical role in strengthening adaptive capacity. Greater participation of vulnerable communities and incorporation of indigenous knowledge can improve the effectiveness and sustainability of adaptation interventions.

Monitoring adaptation outcomes remains a major gap.

Most existing programmes monitor implementation rather than resilience outcomes. Developing measurable adaptation indicators, sectoral resilience metrics and periodic vulnerability assessments will enable the state to evaluate whether adaptation investments are effectively reducing climate risks over time.

Key Resources

This section provides information on additional technical information that is available to support the findings in this report. Information is also provided on future research opportunities